



MILL TEST CERTIFICATE

AXIS PIPE AND TUBE, LLC.
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Customer:
MRC Global
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Sales Order:
0630004951
PO:
S 7AA 697379

Invoice No.:
1600049629
Date of Issue:
07/23/2024

Certificate No.: 1600049629-13
Product Description: SL 12 750 0.500 65.49# X52BDH-SEBW-20P2
Specification: SL 46th Edition May 2019

Heat Number	Lot Number	Mat Number	Steel Marking / Coil Rolling
A402398	L1724185022	332758	Steel Dynamics Columbus, LLC - Columbus - MS - US / Steel Dynamics Columbus, LLC - Columbus - MS - US

Tensile Test - Strip Specimen Gauge Length 2"										Mechanical Properties																			
Loc. / Orient	Size	0.5%EL Y.S.(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)	Temp (F) -20					CVN					Temp (F) -20					Size	Loc. / Orient	Acceptance Criteria (ft-lbf)(AVG)				
								Body (ft-lbf)			Shear Area (%)	Weld (ft-lbf)			Shear Area (%)	Lateral Expansion (in)			Lateral Expansion (in)										
								1	2	3		Avg	1	2		3	Avg	1	2	3	Avg	UB	UB	UB	UB	UB	UB	UB	10x7.5 mm
T180°	1 1/2"	66900	73900	77100	0.91	43.8	80	193	200	192	195	100	100	100	100	193	191	93	159	100	100	100	100	100	100	100	10x7.5 mm	3190	15
								Avg			Avg			Avg			Avg			Avg									
								Lateral Expansion (in)			Lateral Expansion (in)			Lateral Expansion (in)			Lateral Expansion (in)												
								UB			UB			UB			UB			UB			UB						

Analysis																				Chemical Composition (Max %)									
(H/P)		C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEppm (API Max. A.C.)	CE Results										
H		0.06	0.220	0.82	0.010	0.002	0.07	0.07	0.020	0.14	0.003	0.021	0.026	0.015	0.0002	0.0019	0.0064	0.25	0.12										
P		0.08	0.216	0.77	0.011	0.002	0.06	0.07	0.010	0.14	0.003	0.016	0.024	0.014	0.0003	0.0020	0.0050	0.25	0.14										
P		0.07	0.215	0.77	0.011	0.002	0.06	0.07	0.010	0.14	0.003	0.016	0.024	0.014	0.0003	0.0022	0.0051	0.25	0.13										

Following Controls / Tests have been satisfactorily performed:
1. - Hydrostatic Testing: 4,090.00 psi Pressure for 5.00 seconds
2. Ultrasonic Tested to N10 - passed
3. Seam Annealed min 1,600 F
4. Flattening Test Passed

Acronyms:
L = Longitudinal
Y.S. = Yield Strength
E.L. = Elongation
P = Product Analysis
UB = Un-Broken

T = Transverse
U.T.S. = Ultimate Tensile Strength
H = Heat Analysis
A.C. = Acceptance Criteria
N/A = Not Applicable

Remarks:
1: This material meets Grades ASTM A53/A53M-22 Grade B API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 07/2024
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1 B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

John Flayna

Quality Manager